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JEM

A socio-technical systems approach of the accident analysis in Indonesian multiple train accident cases: An application of AcciMap methodology

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ABSTRACT

The number of cases of train accidents in Indonesia continues to be large in the past few years. This paper analyzes train accident cases in Indonesia using a socio-technical system approach accident model called AcciMap, developed by Jens Rasmussen. The source of data comes from train accident investigation reports from 2015 to 2021 published by the National Transportation Safety Committee. This study uses a qualitative approach with thematic analysis to identify contributing factors and their interactions within the reports. Based on the socio-technical system approach, there are roles and contributions from the five levels of the socio-technical railway transportation system in Indonesia that interact with each other in the occurrence of accidents, including the railway regulatory bodies; other organizations; management of railway service company; processes and physical actions by staff; and condition of equipment and environment. This study has shown that the AcciMap methodology can provide a comprehensive view of the problems and their interactions across the socio-technical levels that influence each other in the occurrence of train accidents in Indonesia. The results also suggest that the socio-technical system approach is expected to be applied in the accident investigation process for the related industry to get more comprehensive insights into the accident.

Key words: train accident, accident analysis, socio-technical systems, Indonesia train accident, AcciMap

INTRODUCTION

Train accidents in Indonesia have been a significant concern, with 143 incidents recorded between 2015 and 2021.[1] These cases of train accidents in Indonesia still tend to fluctuate. These accidents resulted in 132 fatalities, 124 severe injuries, and 159 minor injuries.[1] Rail accidents have been observed not only in Indonesia but also in other Asia-Pacific countries. For example, from January 2001 to December 2010, Australia reported 392 fatalities resulting from rail accidents.[2] Likewise, China has experienced a notable frequency of railway accidents.[3] Over the period since 1978, there have been 13 significant railway accidents, each resulting in over 30 fatalities, culminating in a total of 987 deaths. On average, this equates to approximately 29 deaths per year. Given the occurrence of railway accidents on an international scale, this emphasizes the importance of comprehensively addressing rail safety.

Beyond the human toll, there are substantial economic ramifications, as exemplified by the European Railway Agency's assessment, indicating that open-line freight train derailments cost over 200 million Euros annually.[4] This was due to the damage to railroad facilities and infrastructure, and operations. Apart from causing quite large material losses, train accidents can also have an impact on cancelations and delays of other train departures, causing many passengers to switch to other modes of transportation. This may result in reduced public trust in using this form of transportation and a decrease in its popularity among the public. This is due to the bad image of railroad services, which can make people reluctant to use train services because of safety issues.

The extent of our learning relies on the depth of our accident analysis.[5] Thus, there is an imperative

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need to address and prevent the recurrence of similar accidents by comprehensively identifying their underlying contributory factors through the utilization of accident models. Based on statistical data from the Directorate of Railway Safety during that period, the factors causing the most accidents were infrastructure (55 percent), followed by rolling stock (21 percent), operators (11 percent), external (8 percent), and nature (5 percent).

Analyzing the accident phenomenon deeply through accident models is crucial for learning and understanding. Previous research in Indonesia tried to understand train accidents using the Swiss Cheese Model (SCM).[6] Although the SCM is able to analyze the causal factors, this model cannot explain the interactions between factors systemically. Furthermore, the SCM has drawn criticism due to its sequential technique, which oversimplifies accident causation by not considering the complex interaction of system components.[7]

Train accidents can be considered a complex socio-technical phenomenon because they involve various aspects and actors that play a role in the occurrence of accidents.[7,8] On the other hand, the accident investigation process in the transportation sector still focuses a lot on "hunt for the broken component" rather than "hunt for the interacting system components."[8,9] This is considered a significant barrier to improving safety in industries with complex socio-technical systems.[8-10]

To gain a comprehensive understanding of socio-technical systems, the AcciMap method serves as a valuable tool. This methodology, rooted in a system's approach, is extensively employed across diverse industries for accident analysis purposes.[8] AcciMap's approach to accident analysis is underpinned by Jens Rasmussen's risk management framework,[11] which acknowledges accidents as intricate outcomes shaped by decisions, actions, and interactions spanning all levels of a system. This stands in contrast to the conventional practice of attributing accidents solely to frontline personnel or isolated factors. Moreover, AcciMap's approach enables the identification of a network of contributory factors spanning six hierarchical levels, encompassing government and regulatory bodies, local work processes, and the operational environment.[12]

The AcciMap method has gained recognition and is well-established in several industrial sectors, including land,[9,13] sea transportation,[14] and construction.[13] Despite its advantages, it is noteworthy that there is a lack of research utilizing the AcciMap method in Indonesia. Consequently, this study aims to employ the AcciMap method to analyze the factors and their interdependencies contributing to a series of train accidents in Indonesia, addressing this gap in the existing literature.

Thus, this study aims to analyze the factors contributing to train accidents in Indonesia between 2015 and 2021, focusing on the roles of various actors within the rail transportation system. While standard rail accident investigations identify single causes, they often overlook underlying factors and the interaction between those factors, such as safety management issues and regulatory shortcomings within organizations. To address this gap, the current research employs the AcciMap socio-technical systems method to examine how the actions and activities of socio-technical system actors within Indonesia's rail transportation sector have influenced the occurrence of train accidents during the specified period.

METHODOLOGY

AcciMap methodology

AcciMap is a socio-technical accident analysis method developed by Jens Rasmussen based on a risk management framework.[11,15] In this method, there are several levels involved in a system in the organization in carrying out safety risk management starting from the top level, namely, the government and regulators; the middle level, namely, organizational management; and the lowest level, namely, staff and operator activities (Figure 1).[11]

This method was developed based on the premise that dramatic changes from dynamic social interaction have resulted in changes in the industry, especially in terms of risk management due to the very fast rate of technological change, increasing the scale of industrial installations, the very fast development of information and communication technology, as well as the aggressiveness of the industry and an increasingly competitive environment.[11] Jens Rasmussen

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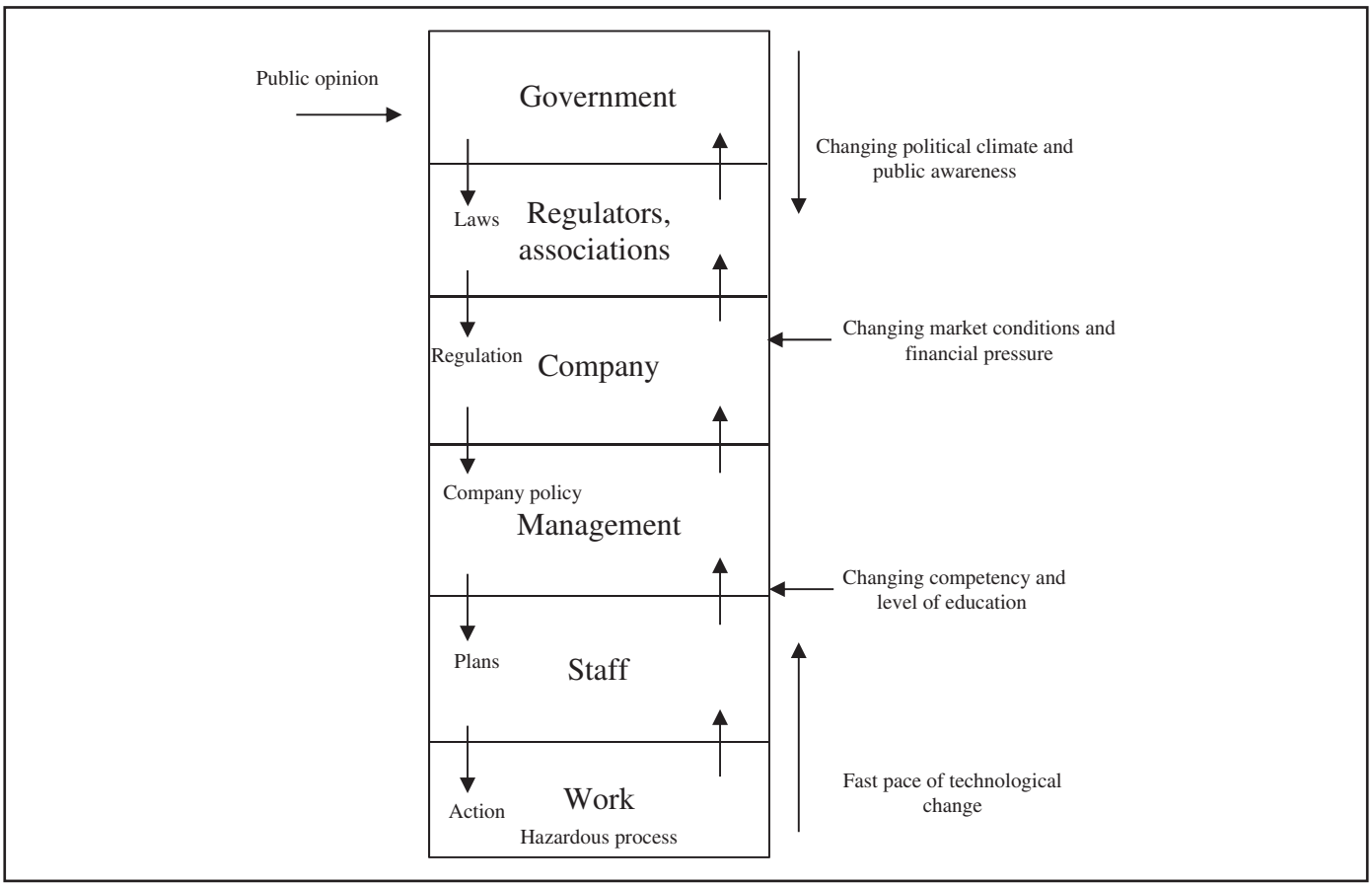

Figure 1. Rasmussen's risk management framework (adapted from Rasmussen[11]).

believes that today's accidents occur due to loss of control of physical processes that can injure people or property damage. Starting from the components of politicians, regulators, directors, managers, safety officers, to frontline workers, all have involvement in controlling it through regulations, policies, procedures, and instructions that are formalized as a form of final control of various hazardous physical processes.[11] Based on this understanding, accidents are caused by not only the interaction of workers at the process level but also due to interactions in the form of decisions and actions made at all levels.

The AcciMap method can describe the contributing factors to accidents mapped at various levels of a complex socio-technical system by relating causes to their consequences both horizontally (within each level) and vertically (between levels). This method can function to identify relevant decision makers and normal work situations where they can influence and create the possibility of accidents, so this method does not focus on finding "who is wrong." AcciMap has been widely applied in analyzing various large-scale organizational accidents in various sectors,[16,17] including land transportation,[9,18] ships,[14] and trains.[7,19]

Data sources

This study uses 30 train accident investigation reports issued by the Indonesia National Transportation Safety Committee (NTSC) from 2015 to 2021 as the unit of analysis. The three types of accidents analyzed in this paper are the most common types of train accidents in Indonesia, namely, derailments, train collisions, and operational disruptions.

Data coding

Analysis in this study used the thematic qualitative analysis method.[9] The contributory factors, actors, and relationships between them were coded

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using Nvivo 11. This process involved a single analyst reviewing each accident report to identify factors—eg, decisions, actions, and/or conditions, and any associated relationships that led to the near miss and accident. To increase the reliability and validity of qualitative analysis, the analysis and coding process was rechecked by two experts and academics in the safety field. If there was a difference of opinion, the final decision was reached through consensus from the discussion.

First, coding are carried out on the contributing factors that are identified from the NTSC report based on the themes that were prepared. For example, a sentence in one of the NTSC reports states, "The derailment of KA 3025 occurred because there was a track twist that exceeded the permissible track twist standard." This statement was given a track twist theme that did not comply with standards so it was given the coding "rail track: track twist fault." Second, the relationship or interaction between contributing factors is marked by arrow symbols. For example, one of the NTSC reports states, "The occurrence of mud pumping has resulted in track twist faults." In this case, mud pumping fault affects the track twist fault. Thus, the arrow in the figure would be "mud pumping" → "tract twist fault." Third, the factors that are identified will be categorized based on the levels of the AcciMap framework. For example, both the "mud pumping" and "track twist fault" are categorized as "Equipment & Environmental Condition." The data coding process is illustrated in Figure 2.

RESULTS

AcciMap framework

Based on the developed AcciMap framework presented in Figure 3, the most common contributing factor themes found in all NTSC investigation reports are training and certification (n = 10), maintenance of facilities and infrastructure (n = 9), problems with the bogie component (n = 8), and track twist fault (n = 8). The following sections will explain in more detail the description of the contributing factors and their interactions based on representative themes at each level of the AcciMap framework that has been created in Figure 3.

Railway regulatory body

Four NTSC reports identified contributing factors at the level of the government or railway transport regulatory body, as shown in Table 1. Three reports stated that there were inadequate policy or regulatory factors issued by the Directorate General of Railways, including procedures for testing facilities and infrastructure being incomplete, lack of competency and proficiency standards for signal house officers, unclear authority rules for train drivers, rules certification of train driver being not comprehensive enough, regulations that are not integrated with other regulations, and definitions and criteria of regulation being unclear. In addition to the inadequate policy and regulatory factors, there was one report that stated that the Director General of Railways did not carry out a safety audit on the company implementing the railway operations.

There are a few statements stating the influence of the actions and activities of the Directorate General of Railways on the levels below. One of them is related to the inadequacy of regulations from the Directorate General of Railways, which affect certification and competency issues at the management level of the company implementing the railway operations. Apart from that, this also affects the level of processes and physical actions, especially those carried out by signaling and telecommunication staff.

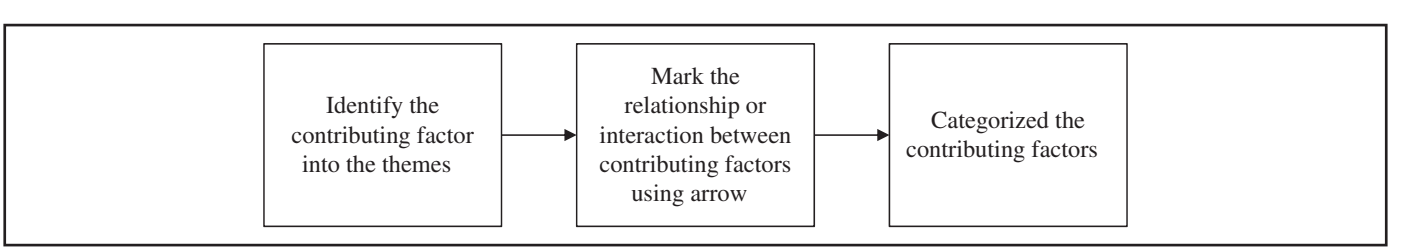

Figure 2. Data coding process.

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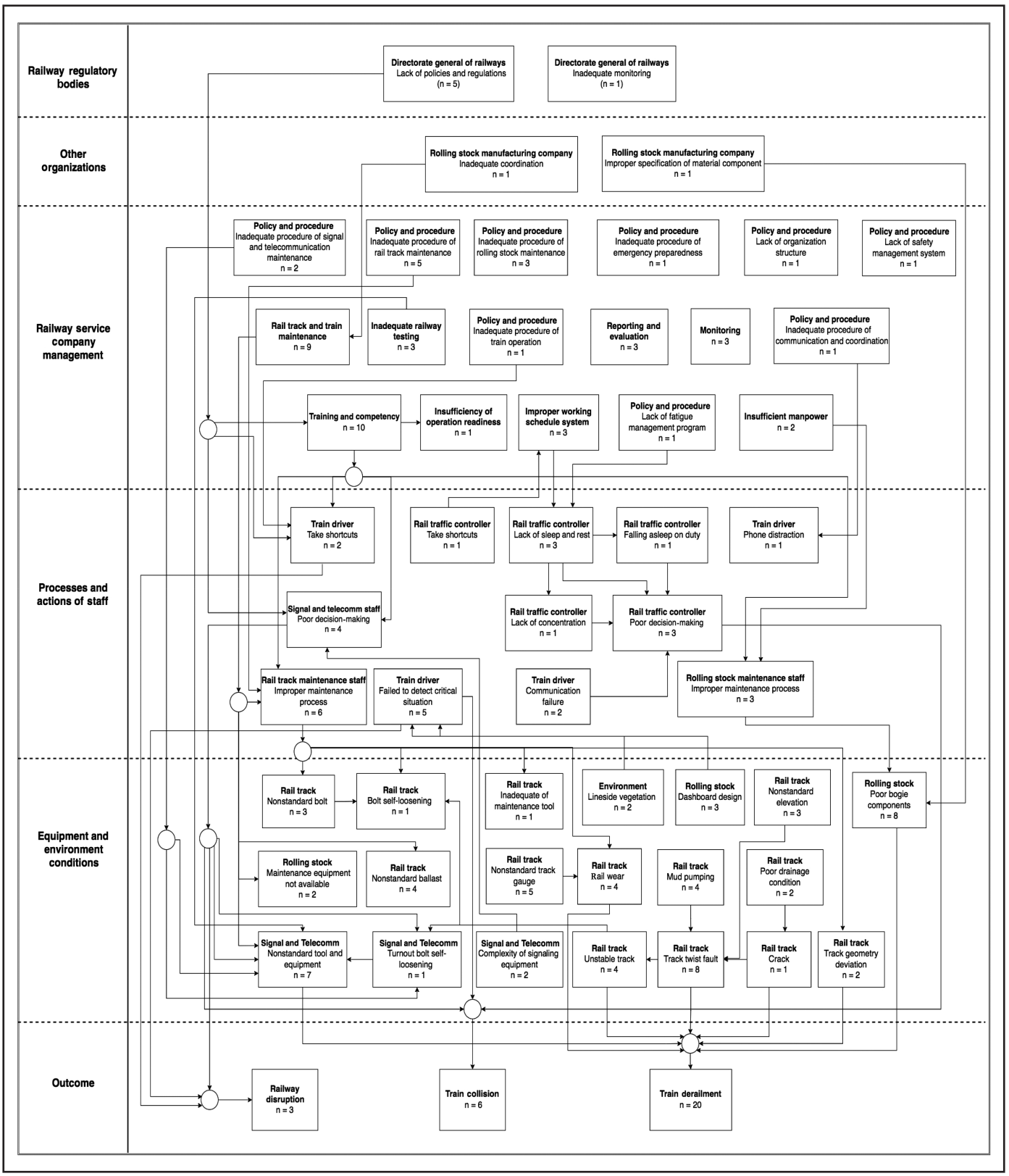

Figure 3. Aggregate AcciMap of the contributing factors, and relationships between those involved in train accidents identified from the NTSC final investigation reports.

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Table 1. Frequency of contributing factors identified by the NTSC underpinning each theme represented on the AcciMap at the "railway regulatory body" level

Contributing factors identified by the NTSC	No.
Policies and regulations by the Directorate General of Railways	1
Lack of rules for testing facilities and infrastructure	1
Lack of standard for competency and skills for signal house officer	1
Unclear train driver authority rules	1
Unclear train driver license and certification rules	1
Regulations are not integrated with other regulations	1
Unclear definitions and criteria	1
Monitoring by Directorate General of Railways	
Lack of safety audit	1

NTSC: National Transportation Safety Committee.

Other organizations

Only one report identified factors at the level of the organization or other related companies. Table 2 shows the contributing factors identified by the NTSC underpinning each theme represented at this level on the AcciMap framework. These two contributing factors have an impact on problems at the lower levels. First, there is an inadequate coordination factor between train operators and outside companies as providers and owners of the Train Control Monitoring System (TCMS) in terms of access to maintenance of facilities, where these factors have an impact on problems implementing maintenance of facilities at the level of companies implementing train operations. Second, the specification of the material component factory of the rolling stock, which is not suitable for the climatic conditions in Indonesia, has resulted in problems with the material for the rolling stock wheels (bogie components) at the level of the condition of the equipment, infrastructure, and the surrounding environment.

Table 2. Frequency of contributing factors identified by the NTSC underpinning each theme represented on the AcciMap at the "other organizations" level

Contributing factors identified by the NTSC	No.
Rolling stock manufacturing company	
Inadequate coordination	1
Improper specification of material component	1

NTSC: National Transportation Safety Committee.

Railway service company management

Twenty-four NTSC reports identified the contributing factors at the management level of the company implementing railway service operations. Most of the relationships and interactions of the contributing factors are identified at this level by illustrating the role of the management of the railway service company contributing directly to the processes and physical actions of train officers, crew, operators, and technicians.

First, 16 reports found problems related to inadequate policies and procedures related to the operation, maintenance, and repair of facilities and infrastructure, some of which had an impact on actions and decisions at the process level and the physical actions of railway company staff and on problems related to the condition of equipment, facilities, and infrastructure.

Second, 10 reports identified problems related to training and certification, which have partially impacted the decisions and actions of train drivers, rail track maintenance personnel, signal and telecom personnel, and rolling stock maintenance personnel.

Third, nine reports identified problems in the maintenance of facilities and infrastructure, some of which have had a direct impact on the process of maintaining the railroad and the availability of equipment for maintenance. Fourth, there are three reports stating factors related to inadequate service schedules, which have resulted in rail traffic controller (RTCs) experiencing a lack of sleep and rest. Finally, two reports stated that the limited workforce that is competent in maintenance results in a failure during the maintenance process (Table 3).

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Table 3. Frequency of contributing factors identified by the NTSC underpinning each theme represented on the AcciMap at the "railway service company management" level

Contributing factors identified by the NTSC	No.
Policies and procedures	
Inadequate procedure of rail track maintenance	5
Inadequate procedure of rolling stock maintenance	3
Inadequate procedure of signal and telecommunication maintenance	2
Inadequate procedure of emergency management	1
Lack of organizational structure	1
Did not implement railway safety management system	1
Inadequate procedure of communication and coordination	1
Inadequate procedure of train operation	1
Lack of fatigue management program	1
Training and competencies	
Lack of rolling stock maintenance staff certification	4
Lack of signal and telecommunication staff certification	2
Inadequate training of rolling stock maintenance staff	2
Inadequate training of train driver	1
Lack of train driver certification	1
Rolling stock and track maintenance	
Inadequate maintenance of facility and equipment	4
Lack of maintenance documentation	3
Poor maintenance quality	2
Inadequate working schedule	3
Insufficient manpower	2
Railway testing	
Lack of signaling equipment test	1

Table 3. Frequency of contributing factors identified by the NTSC underpinning each theme represented on the AcciMap at the "railway service company management" level (continued)

Contributing factors identified by the NTSC	No.
Lack of rolling stock and track test	1
Lack of first and periodic rail track test	1
Reporting and evaluation	
Lack of risk assessment report	1
Lack of corrective action	1
Lack of rail track evaluation	1
Monitoring	
Inadequate monitoring of rail track	1
Inadequate monitoring of procedure implementation	1
Inadequate monitoring of maintenance quality	1
Insufficiency of operation readiness	1

NTSC: National Transportation Safety Committee.

Processes and actions of staff

Twenty NTSC reports identified factors in the processes and physical actions of staff presented in Table 4. Several types of interaction of factors existed at this level. First, there were factors that influence other actions and decisions at the same level. For example, three reports identified RTC who lack sleep and rest. This has resulted in decreased concentration, poor decision-making, and falling asleep while on duty.

Second, there were several actions or factors at this level that affect the management level of the railway company. For example, one report identified the action of changing the service schedule of RTC who were not following the procedure, affecting the levels above, which makes the service schedule not take into account adequate sleep and rest time for RTC while on duty.

Third, there were several factors at this level that affect the level below (equipment and environmental

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Table 4. Frequency of contributing factors identified by the NTSC underpinning each theme represented on the AcciMap at the "processes and actions of staff" level

Contributing factors identified by the NTSC	No.
Train driver	
Failed to detect critical and emergency conditions	5
Failed to communicate with RTCs	2
Taking shortcuts	2
Delay in braking response	1
Using a cell phone while driving	1
RTC	
Lack of sleep and rest	3
Poor decision-making	3
Lack of concentration	1
Taking shortcuts	1
Falling asleep on duty	1
Rail track maintenance staff	
Improper maintenance processes	6
Rolling stock maintenance staff	
Improper maintenance processes	3
Signal and telecommunication staff: Decision-making	
Failed to detect failure when repairing	1
Incorrect service of turnout button	1
Welding process outside the procedure	1
Improper installation of equipment	1

NTSC: National Transportation Safety Committee.
RTC: Rail Traffic Controller.

conditions). Thirteen reports identified contributing factors in the form of inadequate actions by rail track, rolling stock, and signal and telecommunication maintenance staff. Some of these reports stated that these factors influence the condition of quality of bogie components, rail track condition, and signal and telecommunication equipment.

Finally, there were several factors at this level directly related to the occurrence of train accidents. Two reports identified poor decision-making by RTC that had a direct effect on the occurrence of train collision accidents. In addition, one report identified poor decision-making from signaling, and telecommunication staff had a direct effect on the disruption of railway operations.

Equipment and environmental condition

All NTSC reports identified factors at the level of equipment, facilities, infrastructure, and environmental conditions presented in Table 5. The type of interaction of factors that exist at this level is directly related to the same level and the level above it and is directly related to the occurrence of an accident. For example, eight reports identified track twist fault factors. Most of the reports stated that this factor had a direct impact on the unstable rail track and a direct impact on derailments. Another example that shows the influence of the conditions at this level on the levels above is the presence of lineside vegetation that blocks the view and problems with the dashboard cabin design, which resulted in the failure of the train driver in detecting critical situations or emergency conditions.

There were several other factors at this level that were directly related to the occurrence of accidents in addition to problems with the condition of the rail track. Eight reports stated that there were problems with the bogie components at the rolling stock condition including nonstandard wheel components, material damage, and spring suspension problems. Most of the problems at this factor have had a direct impact on the derailment. Seven reports also stated that there was nonstandard signaling equipment, and most of these conditions had a direct impact on the occurrence of the derailment.

DISCUSSIONS

Socio-technical system approach in train accident analysis

This study aims to analyze train accidents in Indonesia from the perspective of a socio-technical systems approach. The case of a train accident can

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Table 5. Frequency of contributing factors identified by the NTSC underpinning each theme represented on the AcciMap at the "equipment and environmental condition" level

Contributing factors identified by the NTSC	No.
Rail track	
Track twist fault	8
Nonstandard track gauge	5
Nonstandard ballast	4
Mud pumping	4
Rail wear	4
Unstable track	4
Nonstandard elevation	3
Nonstandard bolt	3
Poor drainage condition	2
Track geometry deviation	2
Inadequate maintenance equipment	1
Track crack	1
Bolt self-loosening	1
Signaling and telecommunication	
Inadequate signaling equipment	7
The complexity of signaling equipment	2
Turnout bolt self-loosening	2
Rolling stock: Bogie component	
Nonstandard wheel	4
Material damage	2
Spring suspensions	2
Rolling stock: Cabin dashboard design	
Sunvisor blocks the driver's view	1
Barrier on the emergency button	1
The dashboard is difficult to reach	1
Rolling stock: Maintenance facility and equipment	
Limited spare parts	1
Inadequate maintenance equipment	1
Environment	
Lineside vegetation distraction	2

NTSC: National Transportation Safety Committee.

be considered to be a complex system problem. According to Leveson, based on a systems approach, an accident is a condition that arises as a result of nonlinear interactions between various components of a complex socio-technical system.[20] It is the interactions between these components that are important.[20] Based on the socio-technical systems approach, safety is a property of the system that is obtained through the provision of restrictions on the interaction of system components (humans, technology, environment, and so on), which can cause problems in safety control.[11,20,21] To improve the safety aspect and prevent accidents, one of the important things is to analyze the structure, components, and their interactions systematically.

Jens Rasmussen developed a risk management framework for complex socio-technical systems by describing the socio-technical system in a hierarchical structure consisting of several levels from top to bottom consisting of government, regulators and associations, companies, management, staff, and working conditions.[11,22] These levels influence each other through top-down flow, eg, policies, laws, and regulations, and bottom-up information feedback (such as actual system conditions, changes in the external environment, and so forth). The existing levels are not static but dynamic, which will be influenced by the external environment, such as economic conditions, policies, and technological changes.[11]

According to Rasmussen, accidents can occur as a result of failure to control hazardous work processes.[11] The occurrence of control failures is not only caused by the actions or mistakes of certain parties (such as unsafe behavior and actions by workers) but also needs to be seen in a wider socio-technical context to gain more influence and contribution from other levels.[11] According to Salmon and Lenne, the safety of the system is influenced by the decisions and actions of all actors at all levels of the system.[23] Therefore, maintaining system safety is a dynamic control process that involves all levels of the socio-technical system as a whole, and a vertically integrated view of seeing system behavior is necessary.[11,13]

This study used the AcciMap methodology developed by Jens Rasmussen to analyze the socio-technical

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system factors of train accidents in Indonesia. This model can be applied to describe how the decisions, actions, and conditions of various actors in the system interact with each other in creating the occurrence of the accident being analyzed.[9] In addition, in making the AcciMap framework, there is also a process of generalizing the structure and function of the system so that the causality relationship of the contributing factors can be seen, and the system hierarchy can be illustrated through a representative diagram illustration.[13,24,25] Besides being able to describe an overview of complex accident events, the illustrations of AcciMap diagrams can also be the basis for providing a synthesis of assumptions and findings to analyze safety risks and recommendations for future accident prevention.[13]

AcciMap analysis in cases of train accidents in Indonesia

There are several learning points based on the analysis of the AcciMap model framework that was developed in the previous section on the investigation of train accidents in Indonesia. This quite important characteristic of the socio-technical systems approach based on the AcciMap methodology can provide an overview of why and how train accidents in Indonesia occur. The AcciMap methodology has placed various conditions and problems at the levels of the rail transportation system, which interact with each other and then influence the occurrence of train accidents. The characteristics and approach of this model allow unbiased analysis results or only focus on actions and mistakes committed at the frontline level.[14] This is because the AcciMap methodology has succeeded in providing a big-picture perspective and background where events and conditions originate that cause decision and actions taken at the frontline level.[13,14]

For example, based on the developed AcciMap framework in Figure 3, the existence of an inappropriate maintenance process by railroad maintenance officers results because of various other factors at the top level, such as the lack of training from company management, inadequate equipment maintenance facilities, and the absence of procedures (manuals and work instructions) related to maintenance. This has shown that the failure of action on the front line is influenced by many other contributing factors. The factors that have the lowest level, for example, signaling equipment that is not standard, have been influenced by various factors, both factors at the same level and factors at two levels above it.

Evaluation of the NTSC's railway accident investigation

In evaluating the results of a train accident investigation made by the NTSC, this study will compare the results of the AcciMap analysis discussed in the previous session with Jens Rasmussen's predictions regarding the causes of accidents and safety performance.[9,11,13] These predictions are presented in Table 6.

Based on the predictions in Table 6, almost all of Jens Rasmussen's predictions have been fulfilled in the results of the AcciMap analysis, which has been compiled based on the contributing factors and their interactions in all NTSC reports during 2015-2021. However, it can be seen that there is one prediction from Jens Rasmussen, which has not been met based on the results of the AcciMap analysis that has been carried out, that external factors such as economic pressures and political conditions have not been identified, which have a good influence on the actions and decisions of regulatory agencies or the government providing rail transportation as well as the management of the company implementing the operation of the train. This finding shows that although it is quite comprehensive, the investigation process carried out by the NTSC does not fully consider all aspects of the performance of the rail transportation safety system. The results of the analysis that has been carried out highlight that there are two key aspects of the performance of the rail transportation system that have not been identified in the NTSC investigation report.

First, there are no statements in the NTSC reports that explicitly state external influences in the form of economic pressure and political pressure on decisions and actions from a higher level of the system (namely, the government or regulatory bodies and companies implementing railway operations). Most of the reports indicated that the maintenance staff's competence was inadequate due to the lack of

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Table 6. Test of Rasmussen's predictions in train accident

Predictions	Support for prediction
1. Train accident is an emerging property of complex socio-technical systems. It is influenced by the decisions of all actors, be it governments, managers, safety officers, and work planners—not just frontline.	Contributing factors and relationships between factors in the rail transport system have been identified at five levels of the rail safety transport socio-technical system. These factors have uncertain characteristics. In cases of uncertainty, system performance is difficult to predict because individuals adapt to changing demands and conditions. Referring to the AcciMap framework, the performance of the railway transportation system is influenced by a combination of factors that cannot be managed properly. As an example in the case of this study, the failure of the train driver to make the right decision was caused by environmental conditions that obstructed the view and the condition of the cabin equipment (dashboard), which was difficult to reach.
2. Train accident is affected by many contributing factors, not just one decision or action that leads to disaster or catastrophic failure.	The results of the analysis with AcciMap show that various factors that contribute to all levels of the rail transportation system are involved in the occurrence of train accidents. The investigative reports prepared by the NTSC have identified various contributing factors, where most of these factors are also influenced by other contributing factors. For example, the process of maintaining railroads that are not suitable is influenced not only by the inadequate competence of maintenance personnel but also by the availability of incomplete maintenance equipment facilities and the unavailability of procedures for carrying out maintenance.
3. Inadequate railway transport safety performance can result from a lack of vertical integration (lack of effective feedback loops) at all levels of the socio-technical system, not just from deficiencies at one level.	The developed AcciMap framework has identified several examples of nonlinear interactions at all levels of the socio-technical system of rail transport. For example, there is a lot of coordination and feedback seen at the level of "operations management of the railroad operating company" toward the level of "processes and physical actions of officers, crew, operators, and technicians" and the level of "equipment, facilities, infrastructure, and environmental conditions."
4. Inadequate vertical integration is due, in part, to a lack of feedback across levels of the socio-technical system. Actors at each level cannot see how their decisions interact with decisions made by actors at other levels, so threats to safety are not obvious before an accident occurs.	Several examples of poor feedback relationships are identified at various levels of the rail transportation system. One example is that there was poor coordination between the Len Woojin consortium and the airport train manager in accessing TCMS data, resulting in an inadequate maintenance process that causes disruption of train operations. Another example is that the inadequacy of regulations from the Directorate General of Railways regarding the authority of train drivers has an impact on inappropriate operations.
5. Work practices in complex socio-technical systems are not static. It can migrate over time under the influence of various pressures such as financial and psychological pressures.	The developed AcciMap framework showed there are no economic pressure factors that explicitly influence decisions and actions at a higher system level (government or rail transport regulatory bodies).
6. Migration of work practices can occur at various levels of a complex socio-technical system, not just at one level.	The developed AcciMap framework has identified the migration of work practices at various levels. As an example of communication factor, the communication policy at the station, which has not been prepared properly and only relies on cell phones in the group chat platform for coordination and communication, has resulted in the train driver multitasking at the station by operating the train while communicating on cell phones, so that they are not aware of the speed and position of the trains.

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Table 6. Test of Rasmussen's predictions in train accident (continued)

Predictions	Support for prediction
7. Migration of work practices causes the defenses of the system to decrease and erode gradually over time, not all at once.	The developed AcciMap framework describes the mechanisms that produce actions or other decisions in a dynamic work context. Several factors that affect the decrease in system performance appear to decrease over time, for example, the unavailability of railroad maintenance procedures in the field, which, when combined with the lack of competency training or certification for maintenance personnel, as well as the unavailability of adequate maintenance equipment facilities, will create responses and actions that do not comply with standard maintenance practices, which then gradually trigger component conditions. Railroad components are getting worse, which, in turn, causes a decline.

NTSC: National Transportation Safety Committee.
TCMS: Train Control Monitoring System.

competency training and certification for the maintenance staff. In addition, several cases also show the unavailability of complete maintenance facilities and equipment in the field. There may be factors of change and financial conditions at the companies implementing the operation of the railways and the Director General of Railways, which may be one of the external factors that has not been identified in the NTSC report in terms of their influence on contributing factors that influence the lack of the good quality of maintenance of infrastructure and facilities, which has resulted in train derailments and collisions.

Second, there are not many reports that state explicitly the actions, activities, and decisions of regulatory agencies or the government as the highest level in the socio-technical system of rail transportation in Indonesia. Most of the contributing factors identified in the NTSC reports are at the bottom three levels. Out of the 30 reports, only four reports identified as the contributing factors from the Directorate General of Railways. One of the contributing factors identified at this level is inadequate oversight in the form of not carrying out safety audits. Even so, no statement clearly described the effect or impact of not carrying out this audit on other actions. As explained in the previous paragraph, most of the problems at the management level of railway operating companies are related to inadequate training and certification, inadequate policies and procedures, and inadequate maintenance of facilities and infrastructure. These three major problems at the management level of the railway operating company can occur as a result of a lack of oversight from the Directorate General of Railways.

Limitations

This research is only limited based on secondary data sources from investigative reports prepared by the NTSC during 2015-2021. There are likely many other contributing factors, and the relationship between these factors has not been fully identified in the NTSC investigation reports, especially at the upper level of the railway socio-technical system. To obtain more detailed findings for further research, additional data and information can be used, such as through interviews with the NTSC as the accident investigator, the Directorate General of Railways as the regulatory body, PT Kereta Api Indonesia as the company implementing the train operations in Indonesia, and the other parties involved in the operation of the rail transportation system in Indonesia. In addition, further research can also involve practitioners and academics who have experience in the field of rail transportation to confirm the contributing factors that do influence dominantly and accurately the occurrence of train accidents in Indonesia in a comprehensive manner.

CONCLUSION

This study aims to analyze cases of train accidents in Indonesia based on a socio-technical systems approach using Jens Rasmussen's AcciMap methodology.[11] The

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results of the analysis using AcciMap show various conditions and problems as well as their interactions across the levels of the rail transportation socio-technical system in their contribution to train accidents. The interaction of the actions and decisions of various actors influences the conditions and problems of other actors at the same level, the level below it, and the level above it. The results of the analysis using the AcciMap method can provide information for intervention opportunities to improve the safety performance of rail transportation in Indonesia. Therefore, it is necessary to develop and improve the train accident investigation process that takes into account the perspective of a socio-technical systems approach by optimizing information about the role of the top level (government or regulatory bodies) as well as optimizing information related to interactions between factors at various levels, especially at the top levels.

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